

AKSHAY MASTER
SR. DATA ENGINEER

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PROFESSIONAL SUMMARY:

- Overall **8+** years of professional experience as a **Data Engineer**, **Azure**, and **SQL Server** experience in ingestion, data modeling, querying, processing, **storage**, **analysis**, **Data Integration**, and implementing.
- Proficient with **Azure Data Lake Services (ADLS)**, **Databricks** and **python Notebooks** formats, **Azure Databricks Delta lakes** and **Amazon Web Services (AWS)**.
- Experience working with **AWS Cloud services** and **AWS SDKs** to work with services such as **AWS API Gateway**, **AWS Lambda**, **S3**, **AWS IAM**, and **EC2**.
- Proficient in designing and developing **ETL workflows** using **Informatica PowerCenter** to extract, transform, and load data from various sources.
- Skilled in automating data pipelines using **Talend** components to improve data flow efficiency and reduce manual intervention.
- Experience with **NumPy**, **Matplotlib**, **Pandas**, **Seaborn**, and **Matplot Python libraries**. Worked on large datasets by using **PySpark**, **NumPy**, and **pandas**.
- Have experience in **Apache Spark**, **Spark Streaming**, **Spark SQL**, and **NoSQL** databases such as **HBase**, **Cassandra**, and **MongoDB**.
- Knowledge on **Apache Kafka**, **Apache Storm** to build data platforms, **pipelines**, and **storage systems** and search technologies such as **Elastic search**.
- Extensive experience in building scalable big data pipelines using **PySpark** for processing large datasets in distributed.
- Skilled in **Hadoop** daemons such as **Job Tracker**, **Task Tracker**, **Name Node**, **Data Node**, and **MRV** and **YARN** architecture.
- Experience in Installation and Configuring **Hadoop Stack elements MapReduce**, **HDFS**, **Hive**, **Pig**, **Sqoop**, **Flume**, **Oozie**, and **Zookeeper**.
- Experience across both relational databases and non-relational databases **Oracle**, **PL/SQL**, **SQL Server**, **MySQL**, and **DB2**.
- Proficient in database schema design, normalization, and indexing strategies to ensure efficient data retrieval and performance tuning in **PostgreSQL** environments.
- Proficient with complex workflow orchestration tools namely **Oozie**, **Airflow**, **Data pipelines** and **Azure Data Factory**, **CloudFormation** and **Terraforms**.
- Experienced using **Azure Data Factory**, **Airflow**, and **Airflow** on multiple cloud platforms and able to understand the process of leveraging the **Airflow Operators**.
- Skilled in writing complex **SQL queries** and stored procedures to manipulate and transform data within **Snowflake**.
- Expertise in designing and developing interactive dashboards and visualizations using **Tableau** to provide actionable insights from complex datasets for data-driven decision-making.
- Strong skills in visualization tools **Power BI**, **Confidential Excel - formulas**, **Pivot Tables**, **Charts** and **DAX Commands**.
- Experienced in using **Git** for version control of **code**, **scripts**, and configurations across data engineering projects.
- Proficient in using **Jira** for project tracking, task management, and sprint planning within **agile** data engineering teams.

TECHNICAL SKILLS:

Programming Languages	Python, Scala, SQL, R, T-SQL
Clouds	Azure, AWS
Data Processing & ETL Tools	Informatica PowerCenter, Talend, Apache Spark, PySpark, Apache NiFi, AWS Glue, Oozie, Airflow
Databases	PostgreSQL, Oracle, SQL Server, MySQL, DB2, Snowflake, HBase, Cassandra, MongoDB
Data Visualization	Tableau, Power BI, Excel (including advanced features such as DAX, Pivot Tables, Charts)
Big Data Technologies	Hadoop (MapReduce, HDFS, Pig, Hive, Flume, Sqoop, Oozie), Apache Kafka, Apache Flink
Data Analysis Libraries	NumPy, Pandas, Matplotlib, Seaborn
Workflow & Orchestration	Apache Airflow, Oozie, Azure Data Factory, CloudFormation, Terraform
Version Control & CI/CD	Git, GitHub, CI/CD tools, JIRA
Data Architecture	Data ingestion and processing, schema design, normalization, indexing strategies, error handling, logging mechanisms

Scripting & Automation	Python scripts for ETL, Bash scripts, Excel macros, VBA scripts
Security & Compliance	Role-based access control, encryption, audit logging
Project Management	Agile methodologies, JIRA, project tracking and issue management

PROFESSIONAL EXPERIENCE:

Client: Allstate, Northbrook, IL

Jun 2022 – Present

Role: Sr. Data Engineer

Responsibilities:

- Used a combination of **Azure Data Factory**, **T-SQL**, **Spark SQL**, and **U-SQL Azure Data Lake Analytics**.
- Managed **Data** Ingestion to one or more **Azure Services** - **Azure Data Lake**, **Azure Storage**, **Azure SQL**, **Azure DW** and processing the data in **In Azure Databricks**.
- Developed and optimized **ETL** workflows in **Informatica PowerCenter** to extract, transform, and load data from various sources into data warehouses.
- Automated data integration processes using **Informatica** to ensure high-quality data delivery for reporting and analytics.
- Developed and managed **Talend jobs** in production environments, monitoring job performance and resolving issues.
- Created reusable **Talend** components and templates for standardizing **ETL** processes across projects Managed **Snowflake's** virtual warehouses, optimizing performance and minimizing.
- Leveraged **Python libraries** such as **Pandas** and **NumPy** to manipulate large datasets and perform complex data transformations.
- Created reusable **Python modules** for standardizing **ETL** tasks such as data cleaning, validation, and aggregation.
- Developed **Spark scripts**, **UDFS** using both **Spark DSL** and **Spark SQL query** for data aggregation, querying, and writing data back into **RDBMS** through **Sqoop**.
- Ingested data in mini-batches and performs **Spark RDD** transformations on those mini-batches of data by using **Spark Streaming** to perform **streaming Spark analytics** in **Data bricks**.
- Involved in importing real time data to **Hadoop** using **Kafka** and implemented the **Oozie job** for daily imports.
- Involved in the development of real time streaming applications using **PySpark**, **Apache Flink**, **Kafka**, **Hive** on distributed **Hadoop Cluster**.
- Developed **Spark code** using **python** for **PySpark** and **Spark-SQL** for faster testing and processing of data databases to **PostgreSQL**, implementing data transformation and **ETL** pipelines.
- Installed and configured **Hadoop Map Reduce**, **HDFS**, developed multiple **Map Reduce jobs** in **Scala** for data cleaning and preprocessing.
- Created in **Hadoop** architecture and ecosystem such as **HDFS**, **MapReduce**, **Pig**, **Hive**, **Sqoop Flume**, and **Oozie**.
- Worked on **Apache Sqoop** to perform importing and exporting data from **HDFS** to **RDBMS/NoSQL DBs** and vice-versa.
- Worked experience on **NoSQL databases** such as **HBase**, **Azure**, **MongoDB**, and **Cassandra** with functionality and implementation.
- Designed and implemented **NoSQL databases** such as **MongoDB** and **Cassandra** to handle high-volume, unstructured data for large-scale data processing.
- Developed **PostgreSQL** databases for **OLTP** and **OLAP workloads**, facilitating real-time and analytical processing of data across different use cases.
- Created and maintained **DAGs (Directed Acyclic Graphs)** in **Airflow** to ensure reliable and scalable scheduling of batch jobs.
- Designed, developed, and deployed data pipelines using **Apache Airflow** to automate **ETL** workflows for various data processing needs.
- Configured data sharing and replication features in **Snowflake** to facilitate secure and real-time data exchange across different teams and environments.
- Embedded **Power BI** reports on **SharePoint** portal page and managed access of reports and data for individual users using Roles.
- Created a few **Power BI** dashboard reports, **Heat map charts** and supported numerous dashboards, pie charts and heat map charts that were built on Teradata database.
- Integrated **Git** with **CI/CD tools** to automate the build and deployment processes, ensuring rapid and consistent code delivery.
- Used Project Management **services** such as **JIRA** for handling service requests and tracking issues. Valuable experience with **Software methodologies** such as **Agile**

Environment: Azure Data Factory, T-SQL, Spark SQL, U-SQL, Data Lake Analytics, Data Lake, Storage, SQL, Data Warehouse, Databricks, Informatica PowerCenter, Talend, Snowflake, Python, Pandas, NumPy, Spark, Spark DSL, Sqoop, Streaming, Kafka, Oozie, PySpark, Apache Flink, Hive, Hadoop, MapReduce, Scala, Pig, HDFS, Flume, NoSQL, HBase, MongoDB, Cassandra, PostgreSQL, Airflow, Snowflake, Power BI, Git, CI/CD, JIRA, Agile.

Client: Albertsons Companies, Boise, Idaho

Dec 2019 – May 2022

Role: Data Engineer

Responsibilities:

- Worked on Ingesting data by going through cleansing and transformations and leveraging **AWS Lambda**, **AWS Glue** and **AWS Step Functions**.
- Developed and Deployed various **Lambda** functions in **AWS** with in-built **AWS Lambda Libraries** and deployed Lambda Functions in **Scala** with custom Libraries.
- Designed and implemented error handling and logging mechanisms within **Informatica ETL processes**.
- Optimized **Talend** jobs for improved performance, reducing job run times and resource utilization managed Snowflake's resource usage using its built-in monitoring.
- Utilized **Python libraries** such as **Pandas**, **NumPy**, and **PySpark** for data manipulation, analysis, and **large-scale** data transformations.
- Developed **Python scripts** for data extraction, transformation, and loading (**ETL**) **processes**, integrating data from multiple sources.
- Utilized **Spark SQL** for querying large datasets stored in **Hadoop HDFS**, **S3**, and other distributed storage systems.
- Writing the **Spark Core Programs** for processing and cleansing data thereafter load that data into **Hive** or **HBase** for further processing.
- Developed **Spark Streaming programs** to process near real time data from **Kafka**, and process data with both stateless and stateful transformations.
- Integrated **Kafka** with **Spark** and **Hadoop** ecosystems for efficient data ingestion and processing pipelines.
- Implemented scalable and distributed data processing using tools such as **Apache Spark** and **PySpark**.
- Created and maintained technical documentation for launching **Hadoop Clusters** and for executing **Hive** queries and **Pig Scripts**.
- Optimized **NoSQL** database performance by designing efficient schema, leveraging indexing, and tuning query execution for faster read/write operations.
- Implemented robust security measures in **PostgreSQL**, including role-based access control, encryption, and audit logging, ensuring data compliance.
- Integrated **Apache Airflow** with cloud platforms such as **AWS** and manage and orchestrate data pipelines across multiple environments.
- Designed and implemented **ETL/ELT pipelines** for data loading into **Snowflake**, ensuring efficient storage and retrieval of large datasets.
- Created **DAX (Data Analysis Expressions)** formulas and calculated measures in **Power BI** to perform advanced analytics and generate actionable insights.
- Implemented role-based access and security within **Power BI**, ensuring sensitive data is only accessible by authorized users.
- Implemented continuous integration pipelines using **Git** repositories, improving software quality and reducing manual intervention.
- Created and customized **Jira** workflows to **align** with data engineering project needs and team-specific processes.

Environment: AWS Lambda, Glue, Step Functions, Scala, Informatica, Talend, Snowflake, Python, Pandas, NumPy, PySpark, Spark SQL, Hadoop HDFS, S3, Hive, HBase, Spark Core, Streaming, Kafka, Apache Spark, PySpark, Apache Airflow, PostgreSQL, NoSQL, Snowflake, DAX, Power BI, Git, Jira.

Client: Max Healthcare, New Delhi, India

Feb 2017 – Oct 2019

Role: Data Engineer

Responsibilities:

- Integrated both **framework** and **CloudFormation** to automate **Azure** environment creation along with ability to deploy on **Azure**, using build **scripts (Azure CLI)** and automated solutions using terraform.
- Developed an automated process in **Azure cloud**, which can ingest data daily from web **service** and load into **Azure SQL DB**.
- Managed **Informatica** metadata and ensured proper documentation for all **ETL** processes and transformations.
- Developed end-to-end **ETL** solutions using **Talend Data Integration** for various data transformation projects.
- Optimized **Python scripts** for performance, reducing execution times and improving resource efficiency.
- Designed and developed data pipelines using **Apache Spark** for **large-scale** distributed data processing-maintained **Python-based APIs** to expose **data services**.
- Integrated **Kafka** with log aggregation systems to capture and analyze application logs for real-time monitoring and alerting.
- Created on reading and writing multiple data formats such as **JSON**, **Parquet**, and delta from various sources using **PySpark**.
- Installed/Configured/Maintained **Apache Hadoop clusters** for application development based on the requirements.
- Developed data pipelines that integrated **NoSQL** databases with distributed systems, enhancing the scalability and fault tolerance of the overall architecture.

- Worked with **JSONB** and other data types in **PostgreSQL** to handle semi-structured data within a relational environment, enhancing data flexibility.
- Developed custom **Airflow** operators to extend functionality and integrate with third-party tools such as **Spark**, **Hadoop**, and **Snowflake**.
- Developed interactive and visually compelling dashboards in **Tableau** to enable business users to explore and understand data insights effectively.
- Connected **Tableau** to diverse data sources, including **SQL** databases, cloud storage, and **APIs**, to create integrated, real-time reporting solutions.

Environment: Azure, CloudFormation, CLI, Terraform, SQL DB, Informatica, ETL, Talend Data Integration, Python, Apache Spark, Kafka, PySpark, JSON, Parquet, Delta, Apache Hadoop, NoSQL, PostgreSQL, JSONB, Apache Airflow, Tableau, SQL, APIs.

Client: Suryoday Small Finance Bank Ltd, Mumbai, India

Jul 2016 – Jan 2017

Role: Data Analyst

Responsibilities:

- Designed and implemented scalable data pipelines using **AWS services** such as **S3**, **Lambda**, and **EMR** to process large datasets efficiently.
- Wrote **Oozie** scripts and set up workflow using **Apache Oozie** workflow engine for managing and scheduling **Hadoop jobs**.
- Implemented logging, error handling, and monitoring mechanisms in **Python scripts** to ensure smooth production operations.
- Optimized **Tableau** data extracts and live connections for large datasets, ensuring fast performance and minimal downtime in dashboards.
- Managed and optimized **Amazon Redshift** clusters to handle high-volume data storage and retrieval for real-time analytics.
- Automated data extraction and transformation processes by creating **Excel-based macros** and **VBA scripts** to enhance productivity.
- Created detailed technical documentation for data integration pipelines, database schemas, and **ETL processes** using **MS Word**.
- Developed data processing **scripts** in **Python** to automate data extraction, transformation, and loading (ETL) processes from various sources.
- Implemented **Oracle** partitioning strategies to improve the performance and scalability of large data warehouses.
- Developed data models and statistical analyses in **SAS** to uncover business insights and drive decision-making for enterprise clients.
- Implemented **Tableau Server** for report distribution and sharing, setting up user permissions and security for different levels of access.

Environment: AWS S3, Lambda, EMR, Apache Oozie, Python, Tableau, Redshift, Excel, VBA, MS Word, ETL, Oracle, SAS, Tableau Server.

EDUCATION: Jawaharlal Nehru Technology University, Hyderabad, TS, India

BTech in Computer Science and Engineering, June 2012 - May 2016